

Types of functions. Part 2

Logarithmic Functions ⁴ For functions of the form:

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$$f(x) = \log_a(x) \quad \text{or} \quad f(x) = b \cdot \log_a(kx + d) + c$$

where $a > 0$, $a \neq 1$, and $b \neq 0$.

Function Type	Domain	Range
Basic: $f(x) = \log_a(x)$	$(0, \infty)$	$(-\infty, \infty)$
With horizontal shift: $f(x) = \log_a(x - h)$	(h, ∞)	$(-\infty, \infty)$
General: $f(x) = b \cdot \log_a(kx + d) + c$	Solve: $kx + d > 0$ Typically: $(-\frac{d}{k}, \infty)$	$(-\infty, \infty)$

Rational functions For rational functions of the form⁵:

$$f(x) = \frac{P(x)}{Q(x)}$$

where $P(x)$ and $Q(x)$ are polynomials, $Q(x) \neq 0$. A basic example: $f(x) = \frac{1}{x}$

Function	Domain	Range
$f(x) = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$

- **Vertical asymptote** at $x = 0$
- **Horizontal asymptote** at $y = 0$
- **Behavior:** As $x \rightarrow 0^-$, $f(x) \rightarrow -\infty$; as $x \rightarrow 0^+$, $f(x) \rightarrow \infty$

Here are some common patterns and examples

Example	Domain	Range
$f(x) = \frac{1}{x^2}$	$(-\infty, 0) \cup (0, \infty)$	$(0, \infty)$
$f(x) = \frac{x}{x^2 + 1}$	$(-\infty, \infty)$	$[-\frac{1}{2}, \frac{1}{2}]$
$f(x) = \frac{x^2 - 1}{x^2 - 4}$	$x \neq \pm 2$	$(-\infty, \frac{1}{4}] \cup (1, \infty)$
$f(x) = \frac{x - 1}{x^2 - 1}$	$x \neq \pm 1$ (hole at $x = 1$)	$(-\infty, 0) \cup (0, \infty)$

⁵ Finding Domain and Range: Step-by-Step

1. Domain:

- Factor numerator and denominator completely
- Set denominator $Q(x) = 0$
- Domain: All real numbers except solutions to $Q(x) = 0$
- If factors cancel (holes), note them separately

2. Range (requires calculus for certainty):

- Find horizontal asymptotes: Compare degrees of $P(x)$ and $Q(x)$
- Find vertical asymptotes: Zeros of denominator (not cancelled)
- Find critical points: Solve $f'(x) = 0$ or undefined
- Analyze behavior at asymptotes and critical points
- Consider end behavior as $x \rightarrow \pm\infty$

General Rational Functions

Function Type	Domain	Range
Proper Rational (degree of numerator < degree of denominator)	Exclude all x where $Q(x) = 0$ (vertical asymptotes)	Typically all real numbers except horizontal asymptote value, but check with calculus
Improper Rational (degree of numerator $\leq$ degree of denominator)	Exclude all x where $Q(x) = 0$	Typically all real numbers (may have horizontal or oblique asymptote)
With Common Factors (hole in the graph)	Exclude all x where $Q(x) = 0$, including holes	Typically all real numbers except possibly the y -value at the hole location
General Case $f(x) = \frac{P(x)}{Q(x)}$	$\{x \in \mathbb{R} \mid Q(x) \neq 0\}$	Find by: 1. Solving $y = f(x)$ for x 2. Using calculus (derivatives) 3. Analyzing asymptotes and behavior

Calculus Connections for Range Determination

- Use **first derivative test** to find local extrema
- Use **limits at asymptotes** to determine unbounded behavior

$$\lim_{x \rightarrow a^-} f(x) \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) \quad \text{for vertical asymptotes}$$

- Consider **horizontal asymptotes**:
 - If degree of numerator < degree of denominator: $y = 0$
 - If degrees equal: $y = \frac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$
 - If degree of numerator > degree of denominator: No horizontal asymptote (may have oblique asymptote)
- Holes** occur when factors cancel: The y -value at the hole is found by evaluating the simplified function at that x -value

Important Notes for Calculus I

- Rational functions are continuous on their domains
- Domain restrictions affect where derivatives exist
- When integrating rational functions, consider partial fractions for complex denominators
- Always check domain before applying the Intermediate Value Theorem
- For optimization problems, ensure solutions are in the domain